



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

WT901BLECL Bluetooth 5.0 Gyro Sensor



SPECIFICATION

Model : WT901BLECL 5.0

Description : Bluetooth 5.0 AHRS IMU sensor with battery

Quality system standard: ISO9001:2016

Tilt switch production standard: GB/T191SJ 20873-2016

Criterion of detection: GB/T191SJ 20873-2016

Revision date: 2019.12.16

Link to WT901BLECL BLE 5.0 tutorial:(software, manual, etc.):

<https://drive.google.com/open?id=1TWqkdsXo-yMxBGp3Px6CokytSHNwPS1R>



Link to demo video of WT901BLECL BLE 5.0:

https://www.dropbox.com/sh/n5ujs8vswagvqve/AADPVFA_1FA7gAr4aW6T8yMxa?dl=0



Catalog

1	Description.....	- 3 -
2	Product Size.....	- 3 -
3	Features.....	- 4 -
4	Axial Diron.....	- 4 -
5	Method.....	- 5 -
5.1	Connect App.....	- 5 -
5.2	Clibration method.....	- 8 -
5.2.1	Acceleration calibration.....	- 9 -
5.2.2	Magnetic Calibration.....	- 9 -
5.3	Connect to PC software.....	- 11 -
5.3.1	Calibration by Instruction.....	- 15 -
5.4	Restore Factory Setting.....	- 15 -
5.5	Sleep/ Wake up.....	- 15 -
6	Communication Protocol.....	- 15 -
6.1	Module to APP.....	- 16 -
6.1.1	Acceleration, Angular velocity, Angle, Data pack(default)....	- 16 -
6.1.2	Single Return Register Data Packet.....	- 17 -
6.2	APP to Module.....	- 19 -
6.2.1	Read register value.....	- 19 -
6.2.2	Accelerometer Calibration and Magnetic Calibration.....	- 20 -
6.2.3	Save Settings.....	- 20 -
6.2.4	Set Return Rate.....	- 20 -
	Set Port D0.....	- 20 -
	Set Port D1.....	- 21 -
	Set Port D2.....	- 21 -
	Set Port D3.....	- 21 -



6.3	Register address.....	- 21 -
7.	Application Area.....	- 24 -

1 Description

- ◆ Module integrates high-precision gyroscope, accelerometer, mpu9250 geomagnetic sensor, high-performance microprocessors and advanced dynamics solves dynamic Kalman filter algorithm to quickly solve the current real-time movement of the module attitude .
- ◆ The use of advanced digital filtering technology, can effectively reduce the measurement noise and improve measurement accuracy.
- ◆ Integrates gesture solver, with dynamic Kalman filter algorithm, can get the accurate attitude in dynamic environment, attitude measurement precision is up to 0.05 degrees with high stability, performance is even better than some professional Inclinometer!
- ◆ Integrate voltage stabilization circuit, working voltage is 3.3v ~ 5v, pin level compatible 3.3V and 5V embedded system .
- ◆ High-performance cortex-M0 core processor runs at up to 50MHz, taking into account low power consumption and high performance.
- ◆ BLE 5.0 wireless transmission, transmission stability, distance greater than 50 meters.
- ◆ Low power consumption, long standby time, self-contained battery.

- ◆ Stamp hole gold plating process can be embedded in the user's PCB board.

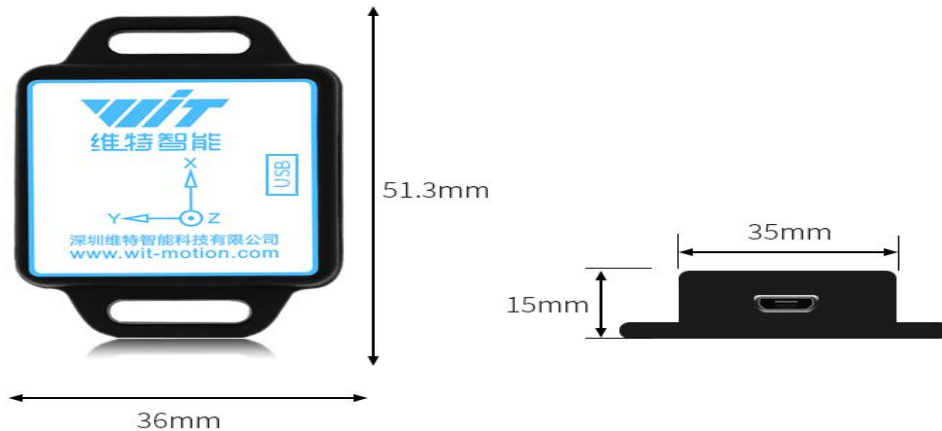
- ◆ 4-layer PCB technology, thinner, smaller, and more reliable.

2 Product Size



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>



3 Features

- 1、 Voltage: 3.3V~5V
- 2、 Working current: $\approx 15\text{mA}$, broadcast current: $\approx 150\text{uA}$, standby current: $10\text{uA}\sim 50\text{uA}$
- 3、 Volume: 51.3 X 36 X 15mm
- 4、 Measuring dimensions:
Acceleration: X Y Z
Angular velocity: X Y Z
Attitude angle: X Y Z
Magnetic field: X Y Z
- 6、 Range: Acceleration: $\pm 16\text{g}$, Angular velocity: $\pm 2000^\circ / \text{s}$, Angle: X Z $\pm 180^\circ$ Y $\pm 90^\circ$
- 7、 Stability: Acceleration: 0.01g, Angular speed $0.05^\circ / \text{s}$.
- 8、 Attitude measurement stable: X Y 0.05° Z : 1°
- 9、 Data output: Acceleration, Angular velocity, Angle, Magnetic field, Port status
- 10、 The data output frequency: 0.1Hz ~50Hz (10Hz default)
- 11、 Data Interface: UART(TTL , Baud rate 115200)
- 12、 Expansion port function: Analog input (0~VCC, Digital input, Digital output)
- 13、 Bluetooth transmission distance: >50m
- 14、 BLE 5.0: Support Android /IOS

4 Axial Diron

As shown in the figure above, the coordinates of the module are indicated, and the upper is the X-axis, the left is Y axis, the Z axis is perpendicular to the surface of the paper to yourself.

The direction of rotation is defined by the right hand rule. that is, the thumb of the right hand is pointed to the axial direction, and the four is the direction of the bending of the right hand.

5 Method

5.1 Connect App

Android App

1. The module builds with a battery, open switch on the side, and open the APP search directly. If you can't find the module, the battery may be out of power. Plug in the data cable and try again.



2. Open the mobile app and click “Scan” to start searching for Bluetooth. The Bluetooth name is “WT901BLE50”.



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

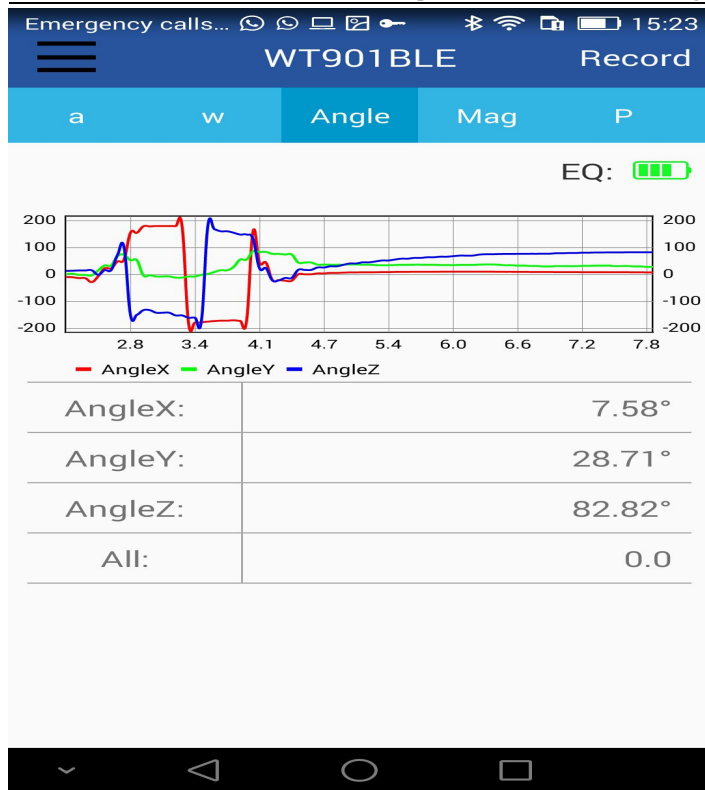


3. Click on the Bluetooth name, the APP cuts into the data display window, and you can see the data returned by the module.



Focus on Attitude Sensor

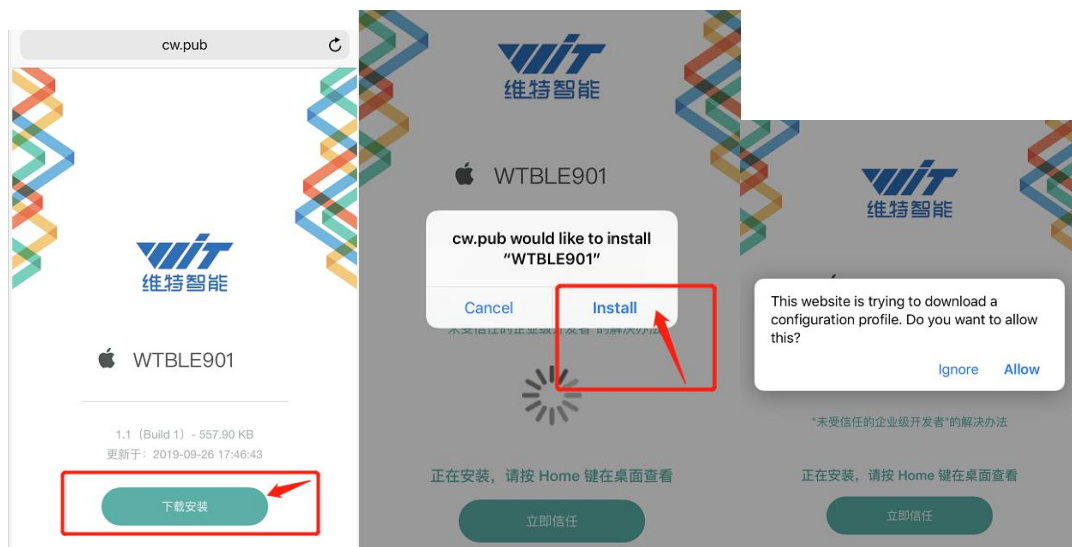
<https://wiki.wit-motion.com/english>



5.1.1 iOS app

Link to app: <https://cw.pub/vrn5>

Step 1: Open link by safari and download the app

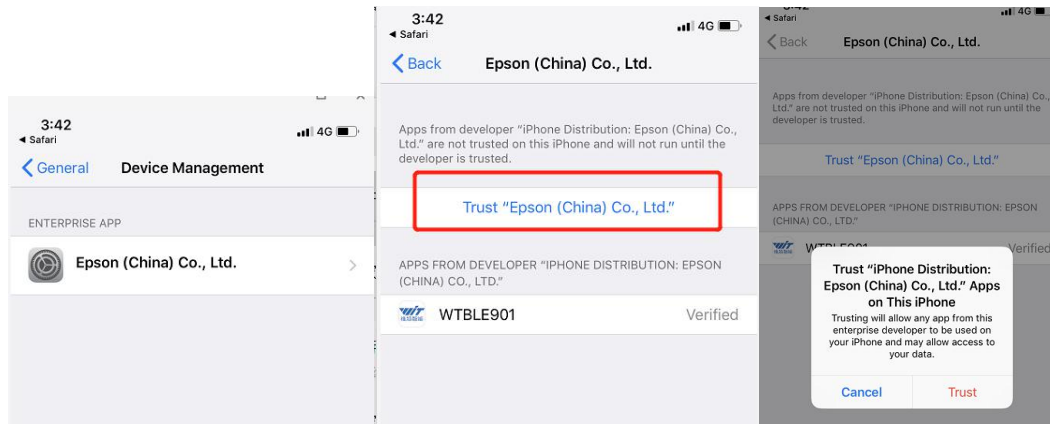




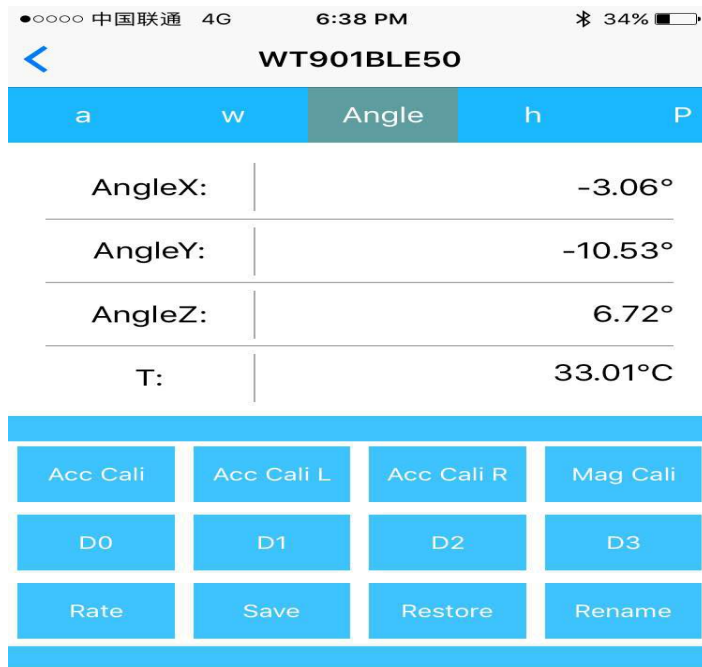
Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

Step 2: Find device management and trust the app launcher.



The connection with IOS app has been shown as below:



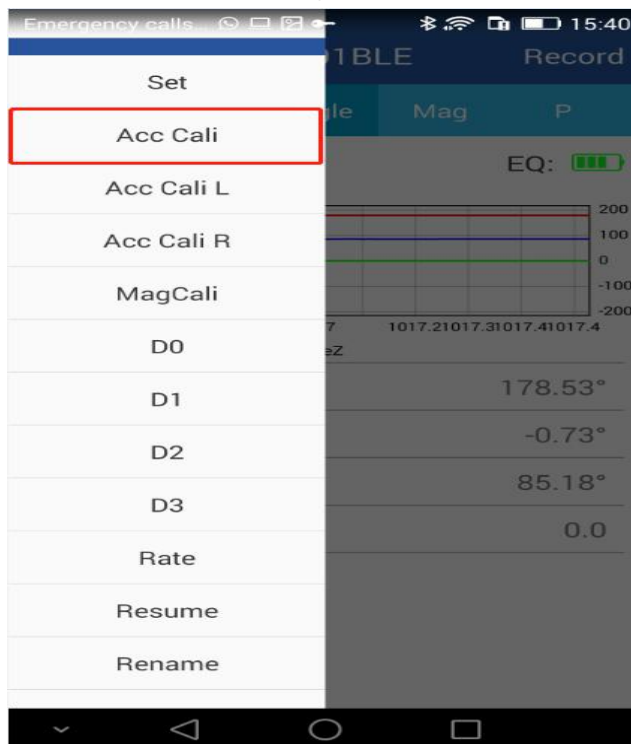
5.2 Clibration method

The module needs to be calibrated before it can be used. Calibration of the module includes additional calibration, magnetic field calibration.

5.2.1 Acceleration calibration

The addition calibration is used to remove the zero offset of the accelerometer. The sensor will have different degrees of zero offset error at the factory, and the measurement will be accurate after manual calibration.

1. To add calibration, the method is as follows: Firstly, keep the module horizontally stationary, click to add calibration, after 1~2 seconds, the three axial values of the module acceleration will be around 0 0 1 and the X and Y axis angles will be around 0°. After calibration, the X Y-axis angle is accurate.



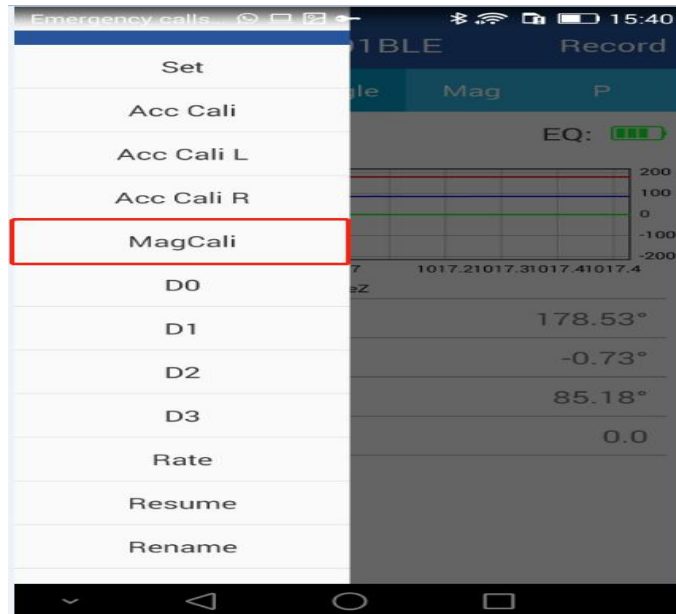
5.2.2 Magnetic Calibration

Magnetic field calibration is used to remove the magnetic field sensor's zero offset. Usually, the magnetic field sensor will have a large zero error when it is manufactured. If it is not calibrated, it will bring about a large measurement error and affect the accuracy of the Z-axis angle measurement of the heading angle.

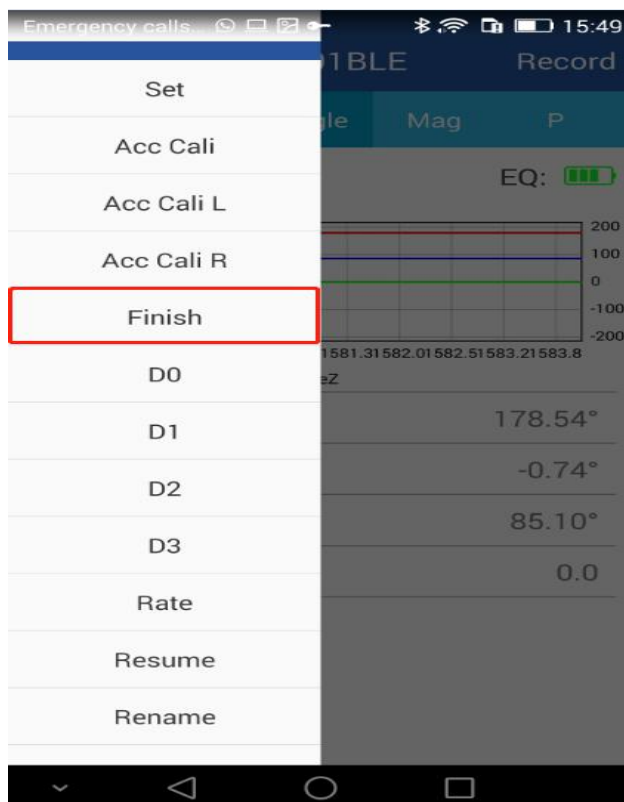
Calibration methods as follow:

1. When calibrating, first connect the module and the computer, and place the module in a place far away from the disturbing magnetic field (ie, more than 20 CM away from magnets and iron, etc.), and then open the upper computer software.
2. Click the “Magnetic Field Calibration” and rotate 360° around the X axis of the module (you can rotate around the Y axis or the Z axis first). Rotate a few turns, then turn 360° around

the Y axis. Then turn 360° around the Z axis, then turn a few turns at random, then click the “Finish” to complete the calibration.



1. Click the “Save” button to save the zero offset data to the internal FLASH of the module. Thereafter, the module is able to output an accurate heading angle.



Note: The data displayed on the APP will not change when the calibration is completed.



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

After the calibration is completed, the data will continue to be transmitted back. When the calibration is added, the module should be stationary. When the magnetic field is calibrated and used, it must be kept away from the magnetic field interference.

5.3 Connect to PC software

if you want to connect this product to software, you should use the special thread which put with product together, use it to connect to the software, we need device CH340 and CP2102 driver to get it.

Driver installation:

First, install the driver CH340 when we used the USB serial module ,after installed the driver. then get the corresponding Com number in the device manager. Driver as followed:

https://wiki.wit-motion.com/english/doku.php?id=communication_module

Resource Summary

User Manual and Development Documents : [communication_module document center](#)
Device driver: [serial_port_debugging_assistant](#) [CH340](#) [CP2102](#)

Contact us

Secondly, install the driver CP2102X when we used the USB serial module ,after installed the driver. then get the corresponding Com number in the device manager. Driver as followed:

https://wiki.wit-motion.com/english/doku.php?id=communication_module

Resource Summary

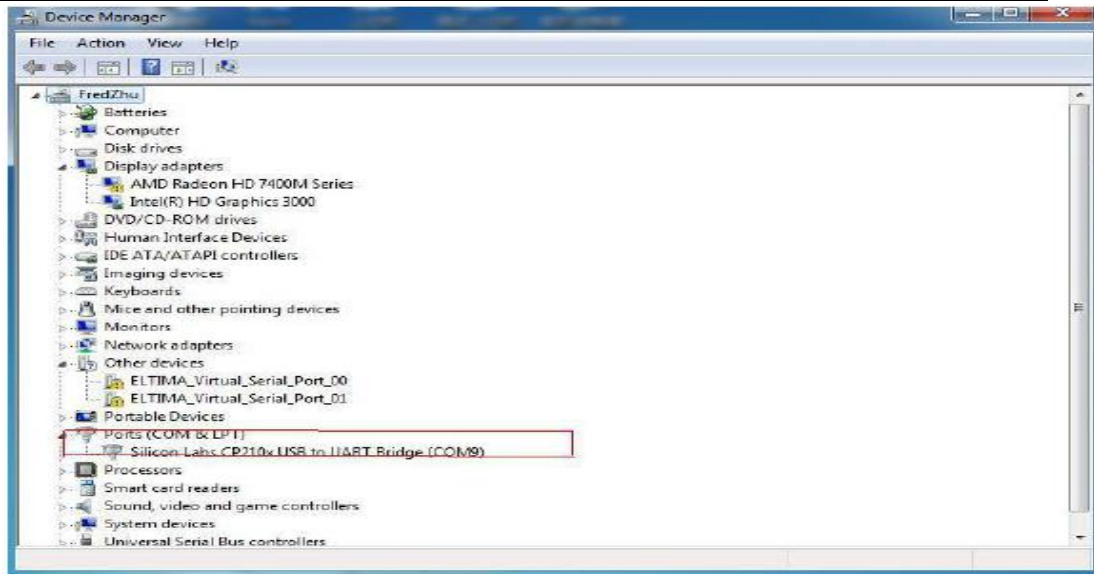
User Manual and Development Documents : [communication_module document center](#)
Device driver: [serial_port_debugging_assistant](#) [CH340](#) [CP2102](#)

After installing the module driver, and then Device Manager can query corresponding serial number, as below figure shows:

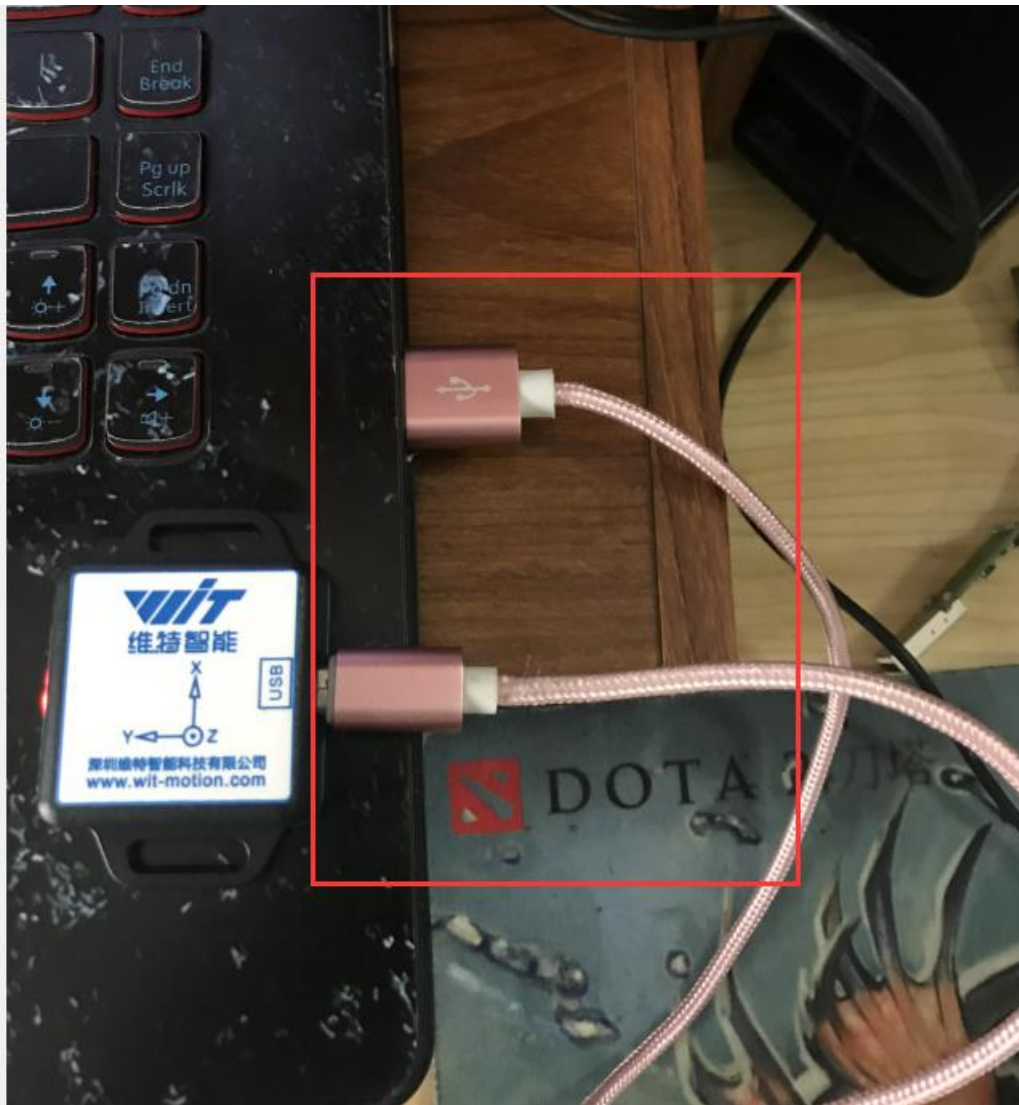


Focus on Attitude Sensor

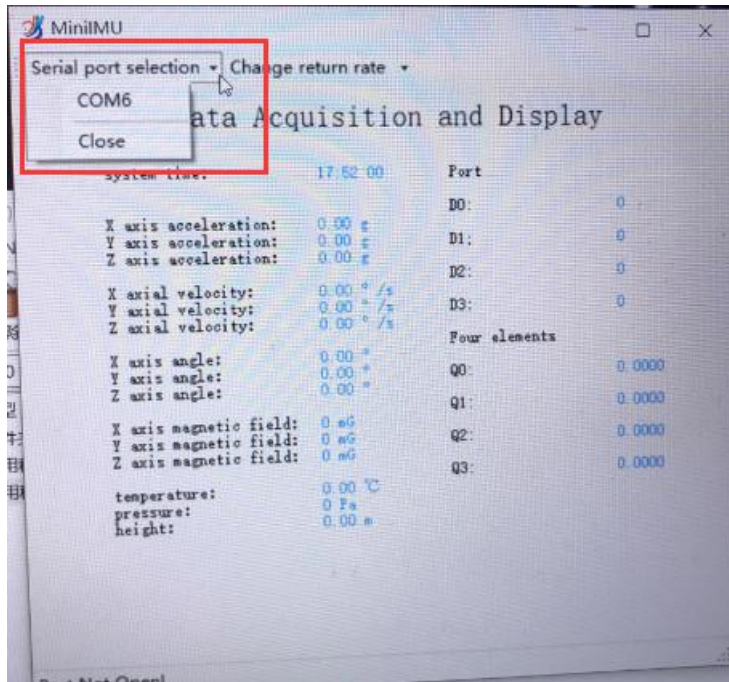
<https://wiki.wit-motion.com/english>



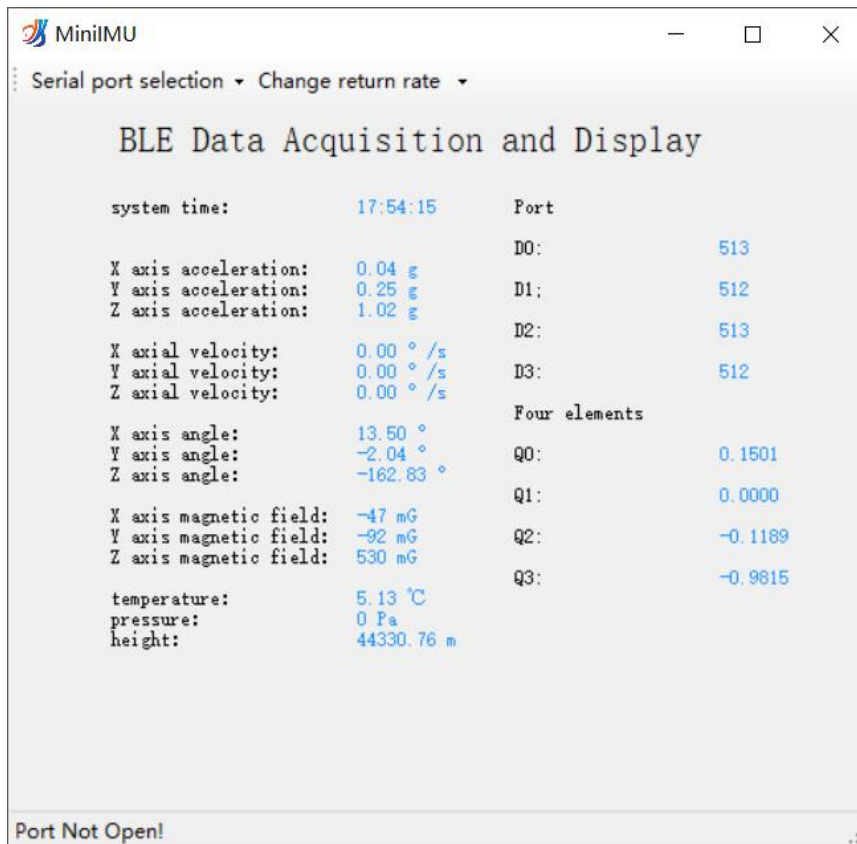
You should use a usb cable to connect to PC software



Then Open the software “MiniIMU.exe” and select the Com number which you have got in the device manager before.



Then the software will show the data





5.3.1 Calibration by Instruction

1. Instruct Accelerometer Calibration:

First keep the module horizontal and still, send the instruction: FF AA 01 01 00, after 1~2s the acceleration X Y Z value will at 0 0 1. X Y angle: 0°. After calibration the value will be accurate.

2. Accelerometer Calibration L/R:

Keep the module is still at left, send the instruction L: FF AA 01 05 00

After 2s, turn the module to the right side and send the instruction R: FF AA 01 06 00

Calibrate two times the data will be accurate.

3. Instruct Magnetic Calibration:

When calibrating, place the module in a place far away from the disturbing magnetic field (ie, more than 20 CM away from magnets and iron, etc.). Send the instruction: FF AA 01 07 00

Rotate 360° around the X axis of the module (you can rotate around the Y axis or the Z axis first). Rotate a few turns, then turn 360° around the Y axis. Then turn 360° around the Z axis, then turn a few turns at random, then click the “Finish” to complete the calibration.

Send the instruction: FF AA 01 00 00 to finish the calibration.

Send the instruction: FF AA 00 00 00 to save the Configuration.

5.4 Restore Factory Setting

Method: After connecting the WT901BLECL and APP via Bluetooth, click the “Resume” .Reconnect the module after recovery.

5.5 Sleep/ Wake up

Enter the sleep mode right two methods. One is to disconnect the Bluetooth connection directly, and the module will go directly to sleep mode. The other is to send a serial port command, the instruction content is 5 hexadecimal data: 0xff 0xaa 0x67 0x01 0x00

There are two ways to wake up the module. One is to directly search for and connect to Bluetooth. The module will wake up automatically and start working. The other is wake-up from the serial port. Any serial port command can be sent to wake up the module.

6 Communication Protocol



6.1 Module to APP

Module upload Flag=0x61 (Angle,Angular velocity, Acceleration) data default.

Flag=0x71(Magnetic field, Air pressure and altitude, Port status) need to send the corresponding register instruction.

Bluetooth upload data: Bluetooth uploads up to 20 bytes per data.

6.1.1 Acceleration, Angular velocity, Angle, Data pack(default)

Packet header 1Byte	Flag bit 1Byte	axL	axH		YawL	YawH
0x55	Flag	0xNN	0xNN		0xNN	0xNN

Note: 0xNN is a accurate value you received. Data return sequence: Acceleration X Y Z
Angular velocity X Y Z Angle X Y Z , low byte first, high byte last.

Flag = 0x61 Data content: 18Byte is Acceleration, Angular velocity, Angle.

0x55	Packet header
0x61	Flag bit
axL	X Acceleration low 8 byte
axH	X Acceleration high 8 byte
ayL	Y Acceleration low 8 byte
ayH	Y Acceleration high 8 byte
azL	Z Acceleration low 8 byte
azH	Z Acceleration high 8 byte
wxL	X Angular velocity low 8 byte
wxH	X Angular velocity high 8 byte
wyL	Y Angular velocity low 8 byte
wyH	Y Angular velocity high 8 byte
wzL	Z Angular velocity low 8 byte
wzH	Z Angular velocity high 8 byte
RollL	X Angle low 8 byte
RollH	X Angle high 8 byte
PitchL	Y Angle low 8 byte
PitchH	Y Angle high 8 byte
YawL	Z Angle low 8 byte
YawH	Z Angle high 8 byte



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

Acceleration calculation method: Unit: g

$$a_x = ((a_{xH} \ll 8) | a_{xL}) / 32768 * 16g \text{ (g is Gravity acceleration, } 9.8\text{m/s}^2\text{)}$$

$$a_y = ((a_{yH} \ll 8) | a_{yL}) / 32768 * 16g \text{ (g is Gravity acceleration, } 9.8\text{m/s}^2\text{)}$$

$$a_z = ((a_{zH} \ll 8) | a_{zL}) / 32768 * 16g \text{ (g is Gravity acceleration, } 9.8\text{m/s}^2\text{)}$$

Calculation method: Unit: °/s

$$w_x = ((w_{xH} \ll 8) | w_{xL}) / 32768 * 2000 \text{ (°/s)}$$

$$w_y = ((w_{yH} \ll 8) | w_{yL}) / 32768 * 2000 \text{ (°/s)}$$

$$w_z = ((w_{zH} \ll 8) | w_{zL}) / 32768 * 2000 \text{ (°/s)}$$

Calculation method: Unit: °

$$\text{Roll (x axis) Roll} = ((\text{RollH} \ll 8) | \text{RollL}) / 32768 * 180 \text{ (°)}$$

$$\text{Pitch (y axis) Pitch} = ((\text{PitchH} \ll 8) | \text{PitchL}) / 32768 * 180 \text{ (°)}$$

$$\text{Yaw angle (z axis) Yaw} = ((\text{YawH} \ll 8) | \text{YawL}) / 32768 * 180 \text{ (°)}$$

Note:

1. Attitude angle use the coordinate system for the Northeast sky coordinate system, the X axis is East, the Y axis is North, Z axis toward sky. Euler coordinate system rotation sequence defined attitude is z-y-x, first rotates around the Z axis. Then, around the Y axis, and then around the X axis.
2. In fact, the rotation sequence is Z-Y-X, the range of pitch angle (Y axis) is only ± 90 degrees, when the pitch angle (Y axis) is bigger than 90 degrees and the pitch angle (Y axis) will become less than 90 degrees. At the same time, the Roll Angle (X axis) will become larger than 180 degree. Please search on Google about more information of Euler angle and attitude information.
3. Since the three axis are coupled, the angle will be independent only when the angle is small. It will be dependent of the three angle when the angle is large when the attitude angle change, such as when the X axis close to 90 degrees, even if the attitude angle around the X axis, Y axis angle will have a big change, which is the inherent characteristics of the Euler angle.

Explanation:

1. The data is sent in hexadecimal format, not ASCII code.
2. Each data is transmitted in descending order of high byte and high byte, and the two are combined into one signed short type of data.

X axis acceleration data Ax: AxL is low byte, AxH is high byte, conversion method as below:

Presume Data is a real data, DataH is high byte, DataL is low byte, so:

$\text{Data} = ((\text{short})\text{DataH} \ll 8) | \text{DataL}$. Please note that DataH need to transform a signed short type of data and then shift. The Data type has a signed short type, so that it display negative number.

6.1.2 Single Return Register Data Packet

Single Return Data Packet need to send register instruction first:



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

FF AA 27 XX 00

--XX is register number. The register number please refer to 7.3. Example as below:

Function	Instruction
Read magnetic field	FF AA 27 3A 00
Read port status	FF AA 27 41 00
Read quaternion	FF AA 27 51 00
Read temperature	FF AA 27 40 00

After send instruction, the module turn back a data packet 0x55 0x71. There are register address and 7 registers data (Fixed upload 8 registers). Return data format as below:

Start register(2 byte) + register data(16 byte, 8 registers)

Packet header	Sign	Start register low byte	Start register high byte	Start(No.1) register data low byte	Start(No.1) register data high byte		No.8 register data low byte	No.8 register data high byte
0x55	0x71	RegL	RegH	0xNN	0xNN		0xNN	0xNN

Note: 0xNN is a accurate value, low byte first, high byte last.

1. Magnetic field output

0x71	0x3A	0x00	HxL	HxH	HyL	HyH	HzL	HzH	..
									

Calculated formular: Unit: mG

Magnetic field (x axis) $H_x = ((H_{xH} \ll 8) | H_{xL})$

Magnetic field (y axis) $H_y = ((H_{yH} \ll 8) | H_{yL})$

Magnetic field (z axis) $H_z = ((H_{zH} \ll 8) | H_{zL})$

EX: Send instruction to read magnetic field in APP: FF AA 27 3A 00 (Please refer to 7.2.8) The module return data to APP: 55 71 3A 00 68 01 69 00 7A 00 00 00 00 00 00 00 00 00 00 00 Total: 20 bytes.

Calculate the no.5 to no.10 bytes as described above, magnetic field x=360, y=105,z=122

2. Port status data output

0x55	0x71	0x41	0x00	D0L	D0H	D1L	D1H	D2L	D2H	D3L	D3H	.
												

Calculated formular:

$D0 = (D0H \ll 8) | D0L$

$D1 = (D1H \ll 8) | D1L$

$D2 = (D2H \ll 8) | D2L$

$D3 = (D3H \ll 8) | D3L$

**Explanation:**

When the port mode is set to analog input, the port status data represents the analog voltage.
The actual voltage is calculated according to the following formula:

$$U = D \times \text{Status} / 1024 \times U_{\text{VCC}}$$

U_{VCC} is chip supply voltage, there is LDO in it, If the module supply voltage $> 3.5\text{V}$, U_{VCC} is 3.3V .

If the module supply voltage $< 3.5\text{V}$, $U_{\text{VCC}} = \text{voltage} - 0.2\text{V}$.

When the port mode is set to digital input, the port status data indicates the digital level status of the port, with a high level of 1 and a low level of 0.

The port status data is 1 when the port mode is set to high output mode.

The port status data bit is 0 when the port mode is set to low output mode.

3 Quaternion output

0x55	0x71	0x51	0x00	Q0L	Q0H	Q1L	Q1H	Q2L	Q2H	Q3L	Q3H	.
												

Calculated formular:

$$Q0 = ((Q0H \ll 8) | Q0L) / 32768$$

$$Q1 = ((Q1H \ll 8) | Q1L) / 32768$$

$$Q2 = ((Q2H \ll 8) | Q2L) / 32768$$

$$Q3 = ((Q3H \ll 8) | Q3L) / 32768$$

Checksum:

$$\text{Sum} = 0x55 + 0x59 + Q0L + Q0H + Q1L + Q1H + Q2L + Q2H + Q3L + Q3H$$

3. Temperature output

0x55	0x71	0x40	0x00	TL	TH	
------	------	------	------	----	----	-------

Calculated formular:

$$T = ((TH \ll 8) | TL) / 100 \text{ } ^\circ\text{C}$$

6.2 APP to Module

Send instruction:

6.2.1 Read register value

FF AA 27 XX 00	Read register value
----------------	---------------------

--XX is register.

EX: Read magnetic field: FF AA 27 3A 00

Read port status: FF AA 27 41 00

Read quaternion: FF AA 27 51 00

Read temperature: FF AA 27 40 00



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

After send instruction, the module turn back a data packet 0x55 0x71. There are register address and 7 registers data (Fixed upload 8 registers). Return data format please refer to 7.1.2.

6.2.2 Accelerometer Calibration and Magnetic Calibration

FF AA 01 01 00	Accelerometer Calibration
FF AA 01 05 00	Accelerometer Calibration L
FF AA 01 06 00	Accelerometer Calibration R
FF AA 01 07 00	Magnetic Calibration
FF AA 01 00 00	Magnetic Calibration Finish

6.2.3 Save Settings

FF AA 00 SAVE 00	Save Settings
------------------	---------------

SAVE: Set

0: Save current configuration

1: Restore default configuration and save

6.2.4 Set Return Rate

FF AA 03 RATE 00	Set return rate
------------------	-----------------

RATE: return rate

0x01: 0.1Hz

0x02: 0.5Hz

0x03: 1Hz

0x04: 2Hz

0x05: 5Hz

0x06: 10Hz (default)

0x07: 20Hz

0x08: 50Hz

0x09: 100Hz

Set Port D0

FF AA 0E D0MODE 00	Set port D0
--------------------	-------------

D0MODE: D0



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

0x00: Analog input (default)
0x01: Digital input
0x02: Output digital high level
0x03: Output digital low level

Set Port D1

FF AA 0F D1MODE 00	Set port D1
--------------------	-------------

D1MODE: D1

0x00: Analog input (default)
0x01: Digital input
0x02: Output digital high level
0x03: Output digital low level

Set Port D2

FF AA 10 D2MODE 00	Set port D2
--------------------	-------------

D2MODE: D2

0x00: Analog input (default)
0x01: Digital input
0x02: Output digital high level
0x03: Output digital low level

Set Port D3

FF AA 11 D3MODE 00	Set port D3
--------------------	-------------

D3MODE: D3

0x00: Analog input (default)
0x01: Digital input
0x02: Output digital high level
0x03: Output digital low level

Note: After the above settings are completed, you must send a save command to save.

6.3 Register address

Address	Symbol	Mean
0x00	SAVE	Save current configuration
0x01	CALSW	Calibration
0x02	KEEP	
0x03	RATE	Return rate



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

0x04	BAUD	UART Baud rate
0x05	AXOFFSET	X Acceleration zero offset
0x06	AYOFFSET	Y Acceleration zero offset
0x07	AZOFFSET	Z Acceleration zero offset
0x08	GXOFFSET	X Angular velocity zero offset
0x09	GYOFFSET	Y Angular velocity zero offset
0x0a	GZOFFSET	Z Angular velocity zero offset
0x0b	HXOFFSET	X Magnetic field zero offset
0x0c	HYOFFSET	Y Magnetic field zero offset
0x0d	HZOFFSET	Z Magnetic field zero offset
0x0e	D0MODE	D0
0x0f	D1MODE	D1
0x10	D2MODE	D2
0x11	D3MODE	D3
0x12	KEEP	
0x13	KEEP	
0x14	KEEP	
0x15	KEEP	
0x16	KEEP	
0x17	KEEP	
0x18	KEEP	
0x19	KEEP	
0x1a	KEEP	
0x1b	KEEP	
.....		
0x30	YYMM	Year, Month
0x31	DDHH	Date, Hour
0x32	MMSS	Minute, Second
0x33	MS	Millisecond
0x34	AX	X Acceleration
0x35	AY	Y Acceleration
0x36	AZ	Z Acceleration
0x37	GX	X Angular velocity
0x38	GY	Y Angular velocity



Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>

0x39	GZ	Z Angular velocity
0x3a	HX	X Magnetic field
0x3b	HY	Y Magnetic field
0x3c	HZ	Z Magnetic field
0x3d	Roll	X Angle
0x3e	Pitch	Y Angle
0x3f	Yaw	Z Angle
0x40	TEMP	Module temperature
0x41	D0Status	D0 Status
0x42	D1Status	D1 Status
0x43	D2Status	D2 Status
0x44	D3Status	D3 Status
0x49	KEEP	
0x4a	KEEP	
0x4b	KEEP	
0x4c	KEEP	
0x4d	KEEP	
0x4e	KEEP	
0x4f	KEEP	
0x50	KEEP	
0x51	Q0	Quaternion Q0
0x52	Q1	Quaternion Q1
0x53	Q2	Quaternion Q2
0x54	Q3	Quaternion Q3

7. Application Area

Agricultural machinery



Internet of things



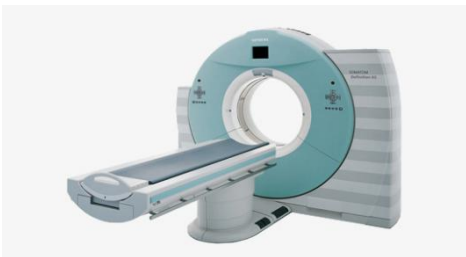
Solar energy



Power monitoring



Medical instruments



Construction machinery





Focus on Attitude Sensor

<https://wiki.wit-motion.com/english>



WT901BLECL BLE 5.0 Bluetooth Accelerometer

Contact: Mr. Kyle Tsang

E-mail : support@wit-motion.com

Skype: live:kyle_8394

WhatsApp: +86 136 523 39539

Amazon in USA: www.amazon.com/witmotion

Amazon in Canada: www.amazon.ca/witmotion

Amazon in Japan: www.amazon.co.jp/witmotion

Official Direct Store: www.aliexpress.com/store/4709011

Address : Honghai building 1405 Songgang town Baoan District

Shenzhen Guangdong Province China